Plugging The $1.67M Annual Revenue Leak

(A Customer Churn & Retention Audit)

Project Overview:

An end-to-end customer retention and revenue protection case study built with Python, MySQL, and Power BI. This project audits customer demographics, contract structures, service bundles, and cancellation reasons to diagnose the root causes of telecom customer churn and help retention teams proactively save at-risk accounts.

Dataset Summary:

Rows: 7043
Columns: 34
Key features:

Customer & Location: [customer_id], [gender], [senior_citizen], [partner], [dependents], [country], [state], [city],[zip_code], [latitude], [longitude].

Billing & accounts: [tenure_months], [contracts], [paperless_billing], [payment_method], [monthly_charges],[total_charges], [tenure_group], [services_taken].

Subscribed services: [phone_service], [multiple_lines], [internet_service], [online_security], [online_backup],[device_protection],[tech_support], [streming_tv],[streaming_movies].

Churn & Risk prediction: [churn_label], [churn_value], [churn_score], [cltv], [churn_reason].

EDA using Python:

I began with cleaning and preparing the data using Python.

Data Loading: Imported dataset using Pandas.

Column Standardization: changed column names to lower case and replaced the space in them with an (_) for better readability and documentation.

Changed [total_charges] column's data type to numeric.

Droped [lat_long] column showing the concatenated value of [longitude] and [latitude] column.

Initial Exploration: Used df.head(), df.info(), df.isnull().sum() to check structure and nulls.

Missing data handling: there were some missing values in [total_charges] column and some empty cells in [churn_reason] for active members.

Filled the [total_charges] nulls with "0.00" and [churn_reason] nulls with "Active Customer".

Handling data impurity: There were some inconsistent values such as ("No phone service" and "No internet service) in [multiple_lines], [online_security], [online_backup], [device_protection], [tech_support], [streaming_tv], [streaming_movies].

I replaced them with "No".

Database integration: Downloaded and set the cleaned Dataframe into MySQL database for further analysis.

Data Analysis using SQL

1. Total number customers vs. Total customers who left us, (%) of customer who left.

total_customers	number_of_churned_customer	customer_churned_rate
7043	1869	26.54

2. Total monthly revenue vs. Total revenue we lost per month, (%) of money lost.

total_monthly_revenue	monthly_revenue_lost	monthly_loss_perc
456116.6	139130.85	30.5

3. Comparison between average monthly revenue contribution of customer who stayed vs. customer who left.

did_they_leave	customer_count	average_monthly_bill
Yes	1869	74.44
No	5174	61.27

4. Average months customers stay before they decide to quit compare to loyal customer.

did_they_leave	average_months_stayed
Yes	17.98
No	37.57

5. Comparison between average lifetime value (CLTV) between customer who left vs. who stay.

did_they_leave	average_lifetime_value
Yes	4149.41
No	4490.92

6. Money lost over a full year because of churn.

total_annual_revenue	annual_revenue_lost	annual_loss_perc
5473399.2	1669570.2	30.5

7. Total lifetime value (CLTV) wiped out by all the quitters.

total_churned_customers	total_lifetime_value_lost
1869	7755256

8. Number of customers quit in their very first 6 months compare to total quitters.

total_quitters	quit_in_first_6_months	first_6_month_quitters_perc
1869	784	41.95

9. How much revenue was lost because of those who quit within first 6 months.

total_churned_customers	monthly_revenue_lost_from_newbies
784	49896.1

10. Top 12 cities with highest quitter %

city	total_customers	total_quitters	total_quitters_perc
Wrightwood	4	4	100.00
Maricopa	4	4	100.00
Tipton	4	4	100.00
Johannesburg	4	4	100.00
Boulder Creek	4	4	100.00
Riverbank	4	4	100.00
Eldridge	4	4	100.00
Smith River	4	4	100.00
Twain	4	4	100.00
Fort Jones	4	4	100.00
South Lake T...	4	4	100.00
Truckee	4	4	100.00

11. Top 10 cities with highest revenue loss per month.

city	monthly_revenue_loss
Los Angeles	6620.7
San Diego	3528.9
San Francisco	2476.6
San Jose	2031.55
Sacramento	1891.2
Fresno	1157.5
Long Beach	1108.1
Oakland	999.25
Stockton	946.35
Modesto	870.2

12. Comparison of quit rate of elderly customers with younger customers.

is_a_senior_citizen	total_customers	total_quitter	quitter_perc
No	5901	1393	23.61
Yes	1142	476	41.68

13. Comparison of quit rate of customers with partner vs. without partner.

have_a_partner	total_customers	total_quitter	quitter_perc
No	3641	1200	32.96
Yes	3402	669	19.66

14. Comparison of quit rate of customers who are senior citizen vs. not senior citizen, with a partner vs. without a partner.

is_a_senior_citizen	have_a_partner	total_customers	total_quitter	quitter_perc
No	No	3072	922	30.01
No	Yes	2829	471	16.65
Yes	No	569	278	48.86
Yes	Yes	573	198	34.55

15. Comparison of quit rate of customers who are dependent vs. not dependent.

is_dependent	total_customer	total_quitters	quitter_perc
No	5416	1763	32.55
Yes	1627	106	6.52

16. Comparison of quit rate of customers with a month-to-month contract who are dependent vs. not dependent.

contract	is_dependent	total_customer	total_quitters	quitter_perc
Month-to-month	No	3239	1549	47.82
Month-to-month	Yes	636	106	16.67

17. Comparison of quitting customers based on internet service.

internet_service	total_customers	total_quitters	quitter_perc
DSL	2421	459	18.96
Fiber optic	3096	1297	41.89
No	1526	113	7.40

18. Comparison of monthly revenue loss based on internet service.

internet_service	total_monthly_revenue	average_monthly_revenue	monthly_revenue_lost
DSL	140665.35	58.1	22529.2
Fiber optic	283284.4	91.5	114300.05
No	32166.85	21.08	2301.6

19. Quitter comparison of Fiber optic users who has tech support vs. who don't.

internet_service	does_have_tech_support	total_user	total_quitters	quitter_perc
Fiber optic	No	2230	1101	49.37
Fiber optic	Yes	866	196	22.63

20. Quitter comparison of Fiber optic users who has online security vs. who don't.

internet_service	does_have_online_security	total_user	total_quitters	quitter_perc
Fiber optic	No	2257	1114	49.36
Fiber optic	Yes	839	183	21.81

21. Quitter comparison of Fiber optic users who has tech support and online security vs. who don't.

internet_service	does_have_tech_support	does_have_online_security	total_user	total_quitters	quitter_perc
Fiber optic	No	No	1765	971	55.01
Fiber optic	Yes	No	492	143	29.07
Fiber optic	No	Yes	465	130	27.96
Fiber optic	Yes	Yes	374	53	14.17

22. Comparison of multiple services users with quitting numbers.

number_of_services_taken	total_customers	customers_quit	quitters_perc
8	208	11	5.29
7	395	49	12.41
6	676	152	22.49
5	908	232	25.55
4	922	289	31.34
3	965	352	36.48
2	1188	390	32.83
1	1701	359	21.11
0	80	35	43.75

23. Comparison on customers monthly bill and lifetime value (CLTV) by number of services they taken.

number_of_services_taken	avg_revenue_contribution_per_month	avg_lifetime_value
8	104.63	5181
7	99.43	5009
6	93.64	4716
5	86.31	4492
4	78.73	4405
3	69.36	4220
2	51.13	4246
1	30.08	4212
0	24.9	4074

24. Comparison of quit rate with each contract type.

contract	total_customer	total_quit	quit_perc	monthly_revenue_lost
Month-to-month	3875	1655	42.71	120847.1
Two year	1695	48	2.83	4165.3
One year	1473	166	11.27	14118.45

25. Comparison of payment methods with quitters.

payment_method	total_customers	total_quit	quit_perc	monthly_revenue_lost
Mailed check	1612	308	19.11	16803.6
Electronic check	2365	1071	45.29	84288.75
Bank transfer (automatic)	1544	258	16.71	20091.9
Credit card (automatic)	1522	232	15.24	17946.6

26. Comparison of different reasons of quitting.

reason_of_quitting	total_quitters
Attitude of support person	192
Competitor offered higher download speeds	189
Competitor offered more data	162
Don't know	154
Competitor made better offer	140
Attitude of service provider	135
Competitor had better devices	130
Network reliability	103
Product dissatisfaction	102
Price too high	98
Service dissatisfaction	89
Lack of self-service on Website	88
Extra data charges	57
Moved	53
Limited range of services	44
Lack of affordable download/upload speed	44
Long distance charges	44
Poor expertise of phone support	20
Poor expertise of online support	19
Deceased	6

27. Group customers based on churn score to check number of customers at risk

risk_status	active_customer_count	revenue_at_risk
low_risk	1679	102240.35
high risk	1675	102699.55
medium risk	1724	105985.5
critical risk	96	6060.35

Dashboard in Power BI

I made an interactable Dashboard to represent insights visually.



Business Recommendation

1. Most of the annual losses comes from month-to-month plans. Offer these customers a discount if they upgrade their plans to one year or two year.
2. Almost half of customers who pay by Electronic Check end up leaving. Offer a discount to those who pays by automatic bank or card payments.
3. Almost half of the new customers quit in their first year. Call new users to take feedback about services and fix them in every 30 to 90 days.
4. Over a 75% of our lost money is due to better competitor deals and rude support staff. Retrain support team and give cheap/free tech support.
5. Fiber optic users pay the highest bill and also quit the fastest. Give them free tech support and online security for their first 6 months.